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THz Image Explorer - An Interactive Cross-Platform Open-Source THz Image Analysis Tool

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Software

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Introduction

THz time-domain spectroscopy (TDS) is a fast-growing field with applications to perform non-destructive studies of material properties (Neu & Schmuttenmaer, 2018). The pulses can either be measured after passing through (transmission spectrum) or after being reflected (reflection spectrum) by a sample and are recorded in the time domain. Through Fourier analysis (FFT), we can investigate the complex refractive index and absorption coefficient of the sample. By placing either the sample or the optical setup on a moving stage the sample can be imaged in 2D.

Figure 1: THz Image Explorer icon.

We developed an interactive graphical user interface (GUI), written in Rust (Matsakis & Klock II, 2014), to aid investigating acquired 2D scans. The application implements the dotTHz standard (Lee et al., 2023) and is platform independent and open-source, thus making it easier to maintain and increasing its reach.

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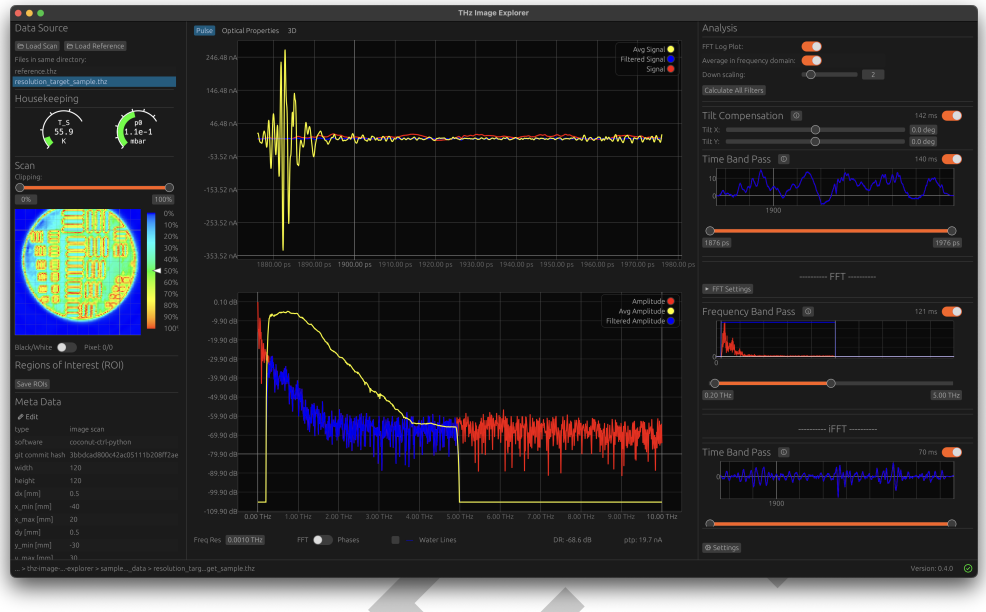

Figure 2: THz Image Explorer screenshot.

Statement of need

Interactive analysis tools for THz spectroscopy are essential to browse through images and analyse different regions of interest efficiently. Commercial suppliers provide closed-source analysis tools (e.g. Menlo Systems) where the code cannot be adapted by the user, which is often essential in research environments and extends the maintainability of the code. Solutions published by the scientific community are not available on all platforms, are only applicable on single pixel measurements and/or not focused on an interactive workflow (Loaiza et al., 2024; Peretti et al., 2019). With this application, we provide a performant solution written in Rust, that allows an interactive analysis of 2D THz scans with multiple filters and a 3D viewer. This work is open-source and pre-built bundles are available for Linux, macOS and Windows, thus available to the entire scientific community.

Structure

The application is multithreaded with two main threads:

- GUI thread
- Data thread

The GUI uses egui, an immediate-mode GUI library for rust.

The structure of the software architecture is shown in figure 3.

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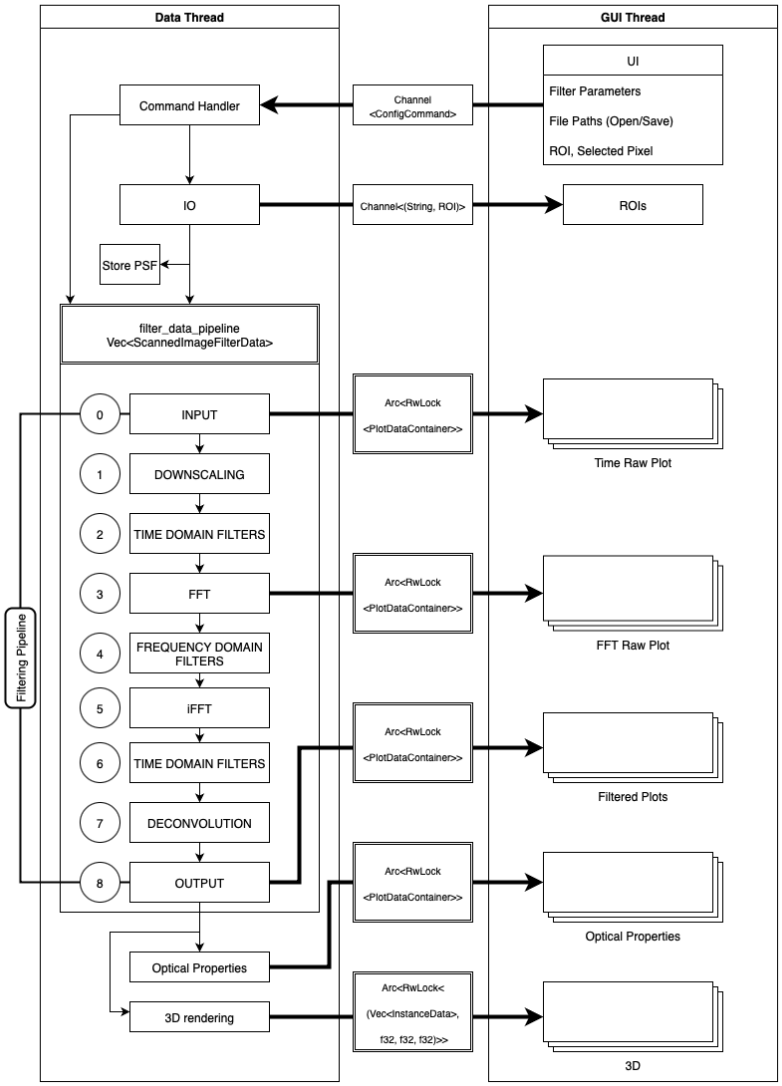

Figure 3: Software Architecture.

Usage

A sample scan (of a resolution target) is available in the `sample_data` directory. The measurement has been acquired using the COCoNuT setup (Stöckli et al., 2025).

Optical Properties Calculation

The refractive index n and absorption coefficient α can be computed using the relations from (Jepsen, 2019). The user can select a source and a reference for the calculation.

$$n(\omega) = 1 + \frac{c\Delta\phi(\omega)}{\omega d}$$

$$\alpha(\omega) = -\frac{2}{d}\ln\left(\frac{(n+1)^2}{4n}\cdot\frac{A_{\text{sample}}(\omega)}{A_{\text{reference}}(\omega)}\right)$$

where:

- $\Delta\phi(\omega) = \phi_{\text{sample}}(\omega) - \phi_{\text{reference}}(\omega)$ is the phase difference
- $A_{\text{sample}}(\omega)$ and $A_{\text{reference}}(\omega)$ are the amplitude spectra
- c is the speed of light
- $\omega = 2\pi f$ is the angular frequency
- d is the sample thickness

Interactive 3D Viewer

A THz time domain scan produces a 3D data array with dimensions $n_x \times n_y \times n_t$, where (n_x, n_y) represent the spatial coordinates and n_t represents the time axis.

Scans performed in reflection can be visualized in 3D. First, we transform each time trace into an intensity value by computing the squared amplitude and applying a Gaussian envelope function:

$$I(x, y, t) = |s(x, y, t)|^2 * G_\sigma(t)$$

where $G_\sigma(t)$ is a normalized 1D Gaussian kernel with standard deviation $\sigma = 6.0$ and radius of 12 samples, applied via convolution to smooth the squared signal and extract the envelope as shon in figure 4.

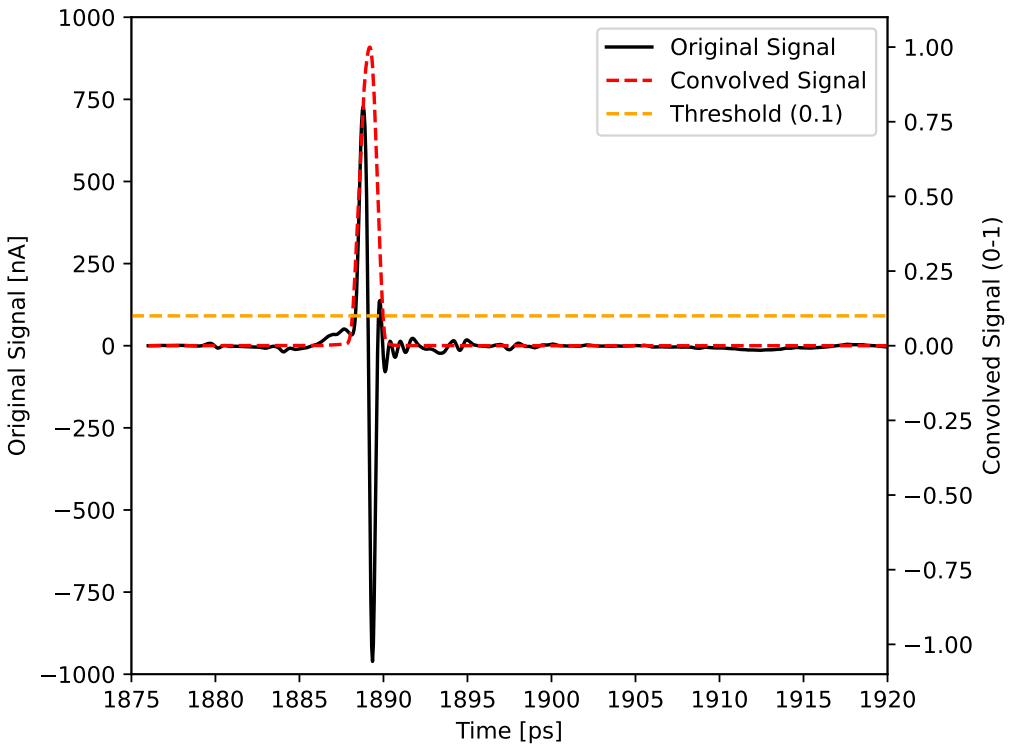

Figure 4: The convoluted envelope of the signal. All datapoints below the indicated threshold are treated as transparent.

The time axis is converted to a spatial distance coordinate by assuming a refractive index of $n = 1$ and using the relation $z = ct/2$, where c is the speed of light and the factor of 2 accounts for the round-trip propagation. This transformation yields a three-dimensional intensity cube $I(x, y, z)$.

Each element (voxel) in this cube represents the THz signal intensity at a specific point in 3D space, enabling visualization of reflections from internal interfaces and sub-surface structures. The computed intensities are mapped to voxel opacity values - regions with high intensity appear opaque while low-intensity regions become transparent.

The 3D viewer is implemented using the `bevy` game engine with a custom WGSL shader, available as a separate crate: bevy_voxel_plot.

Filtering pipeline

The filtering process is a simple linear pipeline, where the output of one filter is the input of the next filter. Filters can be placed in specific domains, either in time domain or frequency domain:

- Time Domain Before FFT
- Frequency Band Pass Filter
- Time Domain After FFT

Band-pass filters are already implemented in the application for each domain. The band-pass filter in time domain before FFT can be used to trim off trailing pulses. By selecting a slice in time domain after FFT, it is possible to scan through the z-axis of the scan and analysing sub-surface layers (Koch-Dandolo et al., 2015).

Deconvolution

The deconvolution filter is an implementation of the Frequency-dependent Richardson-Lucy algorithm described in (Demion et al., 2025).

Custom Filters

The code-base can easily be extended with custom filters. The user needs to create a custom file in the `src/filters` directory with a struct that implements the `Filter` trait.

Summary

THz Image Explorer primarily serves as a performant data analysis tool for THz 2D images. The main focus lies on preliminary browsing of measurements, rough analysis of scans and identifying regions of interest in each scan. It is designed in a modular way to allow possible implementation of more thorough analysis features in the future.

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